LEARN BY BUILDING

LECTURE 14

Transformers and Attention

Context, weighted lookup, and next-token prediction

Language arrives in sequences whose meaning depends on order and context. Attention lets each position combine information from other positions using a learned compatibility rule. A controlled change to one word reveals which context a model can use.

Learning goals. Explain queries, keys, and values; calculate an attention-weighted output; distinguish information access, training, and generation.

1 Why a word needs its context

The same words can assign different roles, and the same word can take different meanings. A useful representation must capture both order and context.

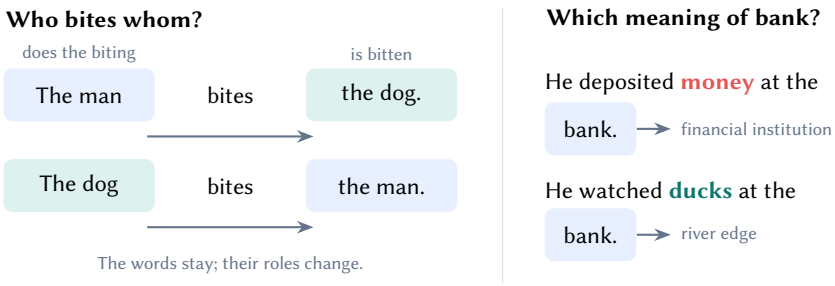


Figure 1. Swapping man and dog reverses who bites whom. Money and ducks suggest different readings of bank. Colors track the words and clues; these are linguistic illustrations.

Tokenization converts text into discrete IDs. A token can be a word, a fragment, punctuation, or a word with a leading space. An embedding maps each ID to a vector. GPT-2 adds a learned position vector, so its initial representation records both identity and position.

Attention lets positions exchange information. A *query* is compared with available *keys*; the resulting weights combine their *values*. These are learned numerical projections. Their roles become concrete in the calculation below.

2 From matching scores to a weighted sum

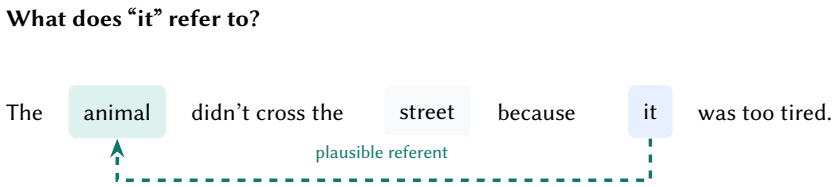


Figure 2. The sentence suggests that the animal was tired. The arrow illustrates a plausible reference; it is not a recorded attention weight or a fixed role assigned to a head.

Let $X \in \mathbb{R}^{T\times d}$ contain the incoming representations for T tokens. One attention head learns three projection matrices:

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V.$$

Queries and keys have width d_k; values have width d_v. The compatibility between query i and key j becomes

$$s_{ij} = \frac{q_i^\top k_j}{\sqrt{d_k}} + M_{ij}, \qquad a_{ij} = \frac{e^{s_{ij}}}{\sum_t e^{s_{it}}}.$$

Softmax acts across the available keys for each query. Its weights are nonnegative and sum to one. Dividing by $\sqrt{d_k}$ controls the growth in dot-product scale with dimension; it is not cosine normalization.

The output is a weighted mixture of *values*:

$$c_i = \sum_j a_{ij} v_j, \qquad C = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V.$$

For a small illustrative calculation, suppose one query has two scores $(\log 3, 0)$. Softmax gives weights $(3/4, 1/4)$. If the values are $(2, -1)$ and $(0, 3)$, then

$$c = \tfrac{3}{4}(2,-1) + \tfrac{1}{4}(0,3) = (1.5, 0).$$

The second coordinate cancels even though both weights are positive. A value coordinate is a signed number; an attention weight is not a percentage of a word’s meaning.

The notebook reconstructs a real GPT-2 head (layer index 5, head index 2): six tokens, 64 value coordinates. Its key operations are

```
scores = Q @ K.T / math.sqrt(d_k)
scores = scores.masked_fill(future_mask, -torch.inf)
weights = scores.softmax(-1)
context = weights @ V
```

The reconstructed output matches the model’s saved head output, with maximum difference zero at displayed precision. Future weights are zero; each row sums to one.

Training learns W_Q, W_K, W_V. The attention weights A are calculated afresh from each input. Confusing these two kinds of weights obscures what attention learns and what changes at inference time.

3 Which words are allowed to contribute?

For causal attention, $M_{ij} = 0$ when $j \le i$ and $M_{ij} = -\infty$ when $j > i$. A token can use itself and earlier positions, but not later ones. The restriction applies at every block.

Only the later word changes in the sentence pair below; *bank* keeps the same identity and position. Its GPT-2 representation stays identical. BERT allows context from both sides, so its representation can change.

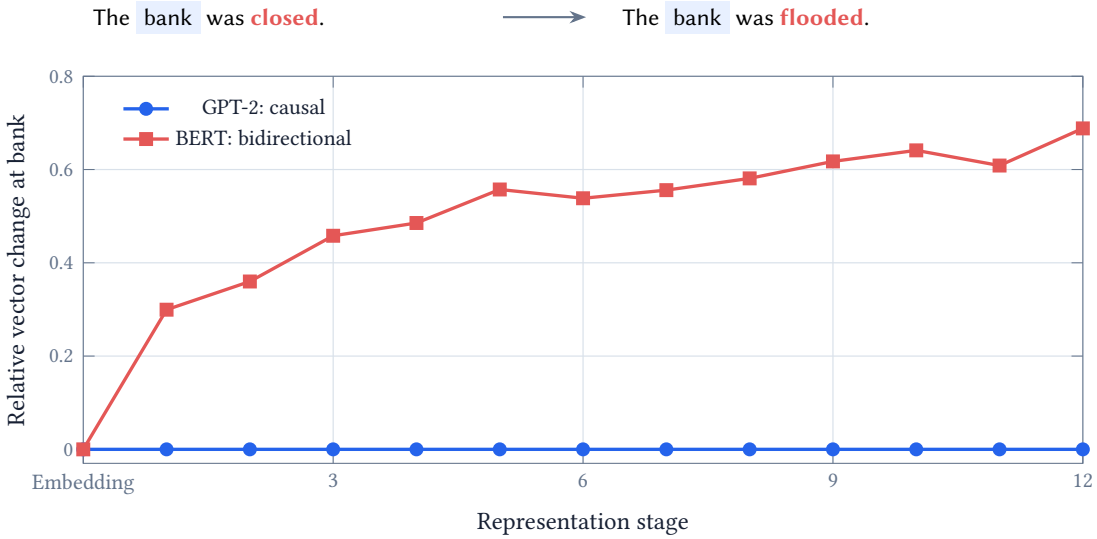


Figure 3. Measured changes to the representation of bank after replacing the later word closed with flooded. GPT-2 stays at zero; BERT changes. The curves use complete hidden vectors, not a two-dimensional projection.

The vertical axis is a within-model relative change,

$$\Delta = \frac{\|h_A - h_B\|_2}{(\|h_A\|_2 + \|h_B\|_2)/2}.$$

A larger value is not an accuracy score or proof of better understanding. This experiment tests information access. The unchanged starting vectors confirm that the target token and its position stayed fixed.

Now change the *earlier* context: “He deposited money at the bank” versus “He watched ducks at the bank.” GPT-2 starts with the same embedding for the final token, then produces different final representations (relative change about 0.138) and different next-token distributions. The causal mask permits that earlier context to reach the target.

Without position information, unrestricted self-attention is equivariant to a permutation of its inputs: it follows the reordered vectors rather than supplying an intrinsic word order. GPT-2 combines learned position embeddings with causal access. Standard BERT instead learns a masked-token objective with access to both sides.

4 From attention heads to a language model

Multiple heads use separate learned projections. Their outputs are concatenated and projected back to the model width. GPT-2 combines 12 heads of 64 coordinates each. Heads are not independent classifiers voting on a word.

Different attention patterns do not establish fixed linguistic roles or causal importance to the final prediction.

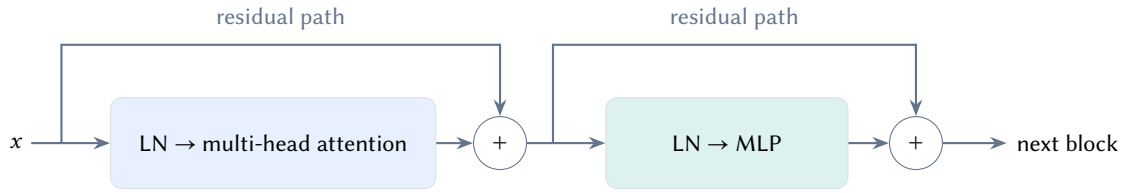


Figure 4. A GPT-2 block mixes information across positions with attention, then transforms each position with an MLP. Residual additions preserve a path for the incoming representation. LN denotes layer normalization.

The MLP shares parameters across positions within a block; different blocks have different parameters. All these paths jointly determine the next representation.

5 Training predicts recorded tokens; generation extends them

A final normalization and vocabulary projection produce logits. Softmax gives a distribution over 50,257 possible next tokens, not a probability that a statement is true.

For recorded tokens $x_1, \ldots, x_T$, next-token training minimizes

$$L = -\frac{1}{T-1} \sum_{t=1}^{T-1} \log p_\theta(x_{t+1} \mid x_{1:t}).$$

The notebook aligns `logits[:, :-1]` with `input_ids[:, 1:]`. All recorded prefixes are available, so causal attention can compute these predictions in parallel. Backpropagation trains the projection matrices without labels specifying which token a head should attend to.

During ordinary autoregressive generation, a selected token is appended before the next prediction. For the money prompt, the saved greedy run selects *bank* at about 23% probability, then a comma, then *but*. A top-ranked token need not have high probability; greedy decoding does not optimize the whole sequence.

Vaswani et al., Attention Is All You Need and Devlin et al., BERT provide the architectural and training context.

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In one sentence: Attention learns input-dependent mixtures of value vectors, while position information, masks, and the training objective determine how those mixtures support sequence prediction.